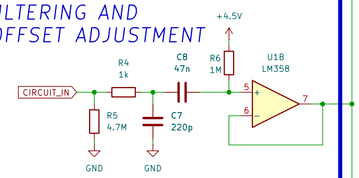
INPUT STAGE: PREFILTERING AND DC-OFFSET ADJUSTMENT

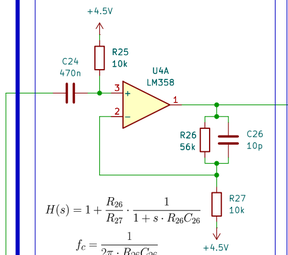


R5: weak pulldown. Pulls the input to ground when no external signal is present.
R4–C7: Passive, Low-Pass filter. Removes unwanted high-frequency noise.
C8: Removes any unwanted DC offset from the input signal.
R6: weak pullup. Injects a stable 4.5V DC offset to the signal
U1B: Voltage buffer. Acts as a low-impedance voltage source for the next stages.

NOISE GATE

NON-INVERTING PRE-AMPLIFIER WITH LOW-PASS FILTER

U4A acts as a non-inverting amplifier when the input is well below the cutoff frequency of the feedback network. When the input is above the cutoff frequency, it acts as a voltage buffer.



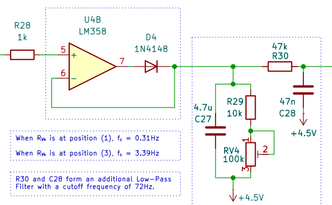
$$H(s) = 1 + \frac{R_{26}}{R_{27}} \cdot \frac{1}{1 + s \cdot R_{26} C_{26}}$$

$$f_c = \frac{1}{2\pi \cdot R_{26} C_{26}}$$

The DC gain of the pre-amplifier is 16.4dB
The cutoff frequency is 284kHz

ENVELOPE DETECTOR

U4B and D4 work together as an ideal diode (ie near-zero threshold voltage)
R29, RV4 and C27 form the Low Pass Filter of the envelope-detector.
The cutoff frequency can be adjusted through the RV4 potentiometer.

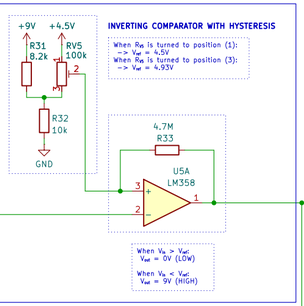


When R_{RV} is at position (1), f_c = 0.31Hz

When R_{RV} is at position (3), f_c = 3.39Hz

R30 and C28 form an additional Low-Pass Filter with a cutoff frequency of 72Hz.

INVERTING COMPARATOR WITH HYSTERESIS

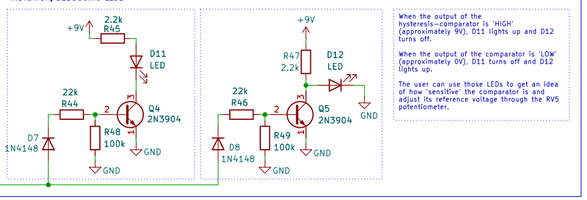


When R_{RV} is turned to position (1):
-> V_{ref} = 4.5V
When R_{RV} is turned to position (3):
-> V_{ref} = 4.93V

When $V_{in} > V_{ref}$:
V_{out} = 0V (LOW)

When $V_{in} < V_{ref}$:
V_{out} = 9V (HIGH)

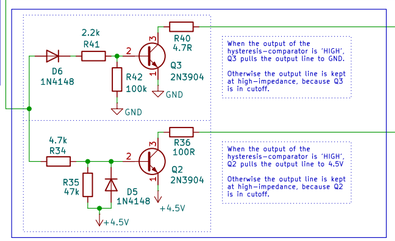
INDICATOR/DEBUGGING LEDS



When the output of the hysteresis-comparator is 'HIGH' (approximately 9V), D11 lights up and D12 turns off.

When the output of the comparator is 'LOW' (approximately 0V), D11 turns off and D12 lights up.

The user can use those LEDs to get an idea of how 'sensitive' the comparator is and adjust its reference voltage through the RV5 potentiometer.



When the output of the hysteresis-comparator is 'HIGH', Q3 pulls the output line to GND.

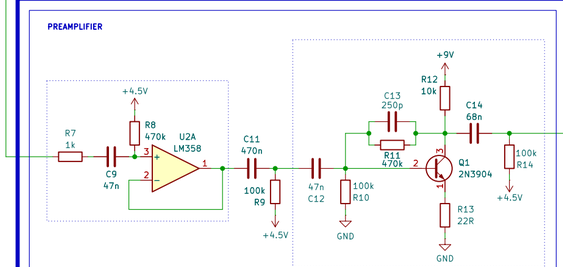
Otherwise the output line is kept at high-impedance, because Q3 is in cutoff.

When the output of the hysteresis-comparator is 'HIGH', Q2 pulls the output line to 4.5V

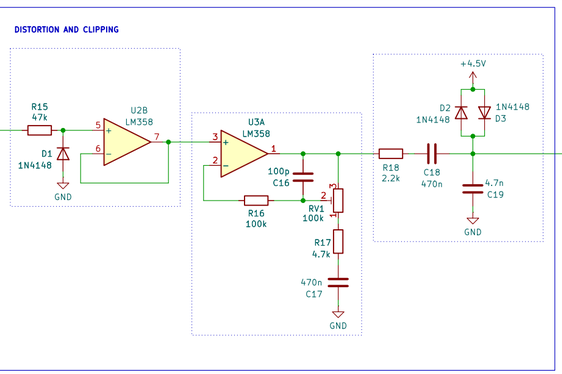
Otherwise the output line is kept at high-impedance, because Q2 is in cutoff.

DISTORTION AND CLIPPING

PREAMPLIFIER

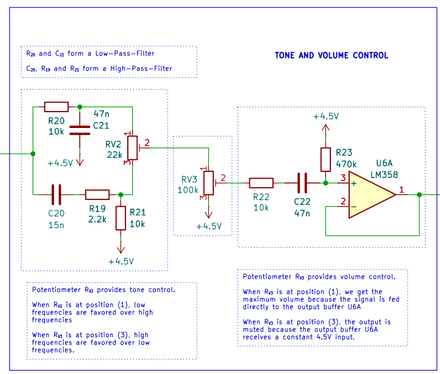


DISTORTION AND CLIPPING



TONE AND VOLUME CONTROL

R_{RV} and C_R form a Low-Pass-Filter
C_{RV}, R_{RV} and R_R form a High-Pass-Filter



Potentiometer R_{RV} provides tone control.

When R_{RV} is at position (1), low frequencies are favored over high frequencies

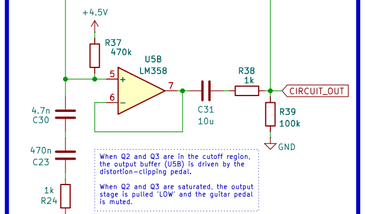
When R_{RV} is at position (3), high frequencies are favored over low frequencies.

Potentiometer R_{RV} provides volume control.

When R_{RV} is at position (1), we get the maximum volume because the signal is fed directly to the output buffer U6A

When R_{RV} is at position (3), the output is muted because the output buffer U6A receives a constant 4.5V input.

OUTPUT STAGE

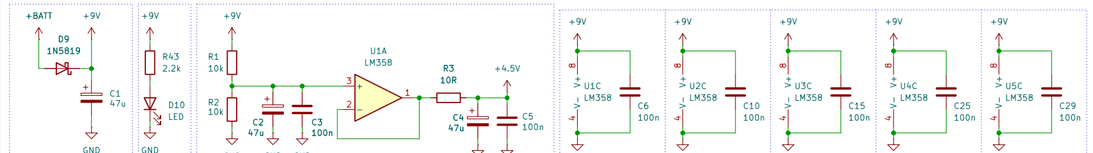


When Q2 and Q3 are in the cutoff region, the output buffer (U5B) is driven by the distortion-clipping pedal.

When Q2 and Q3 are saturated, the output stage is pulled 'LOW' and the guitar pedal is muted.

POWER-SUPPLY: DECOUPLING CAPACITORS AND 4.5V RAIL

D9: Reverse-polarity-protection Schottky diode for 9V battery input.
C1: Decoupling capacitor for 9V rail
D10 – R43: Power indicator LED with current limiting resistor
R1 – R2: Resistor network for generating the reference voltage of the 4.5V rail
U1A: Voltage buffer which acts as a low-impedance 4.5V voltage source.
C4 – C5: Decoupling capacitors for the 4.5V rail.



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